

Thesis of Computer Information Systems & Cybersecurity



AEGISFA: AI-POWERED FORENSIC ANALYSIS FOR SECURITY OPERATIONS

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Problem Statement: Security Teams are Drowning in Log Data

Alert Overload

1,876+ attacks per
quarter (detected)
50% (undetected)



Slow Response

60 min initial triage
Average resolution
time: 239 days



Skill Gap

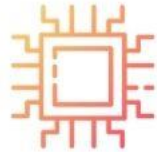
72% of teams lack
expertise and
workers



Solution: Four-Phase Threat Intelligence Pipeline

ML Classification

Automated **Pattern Recognition**
25+ Attack Types Detected



LLM Threat Analysis

Natural Language Insights
Explains **"Why"**



Correlation Engine

Connects **Isolated** Events
Reveals **Full Attack Chain**

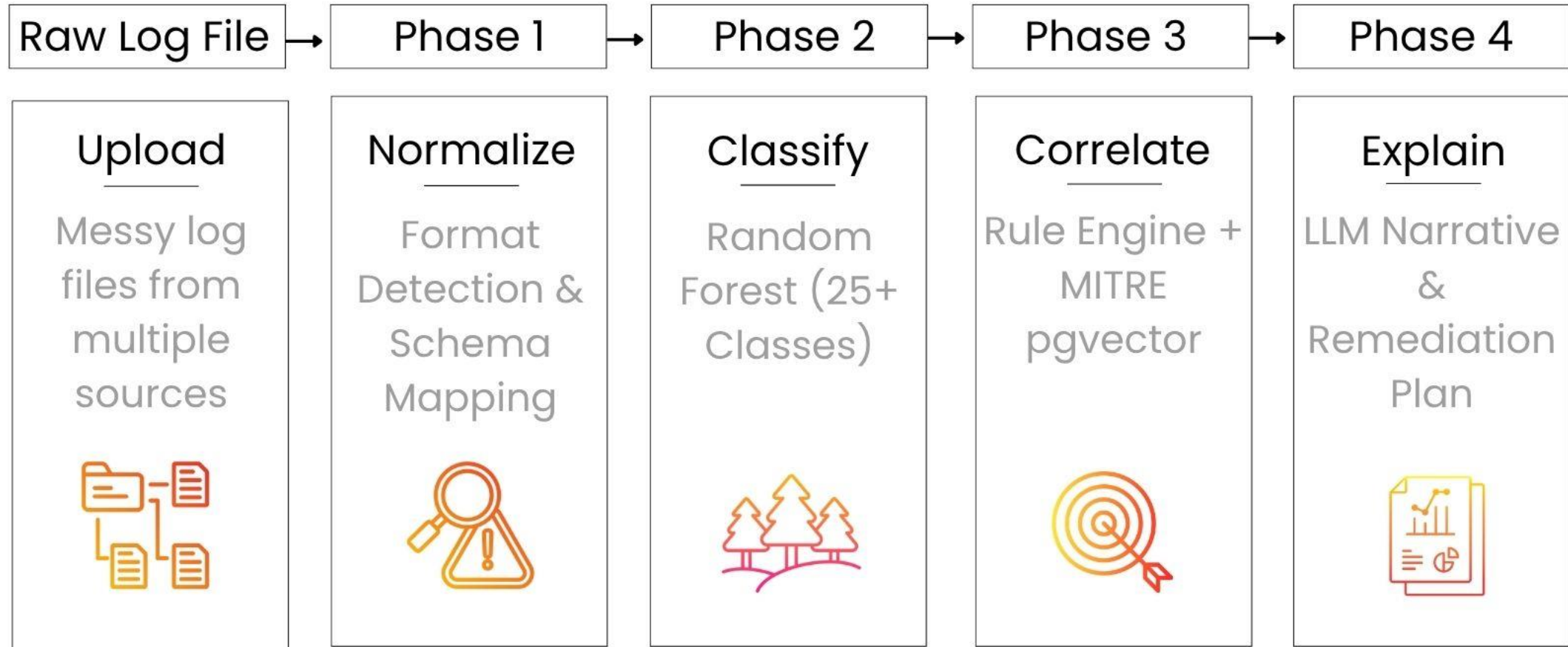


RAG + MITRE Mapping

Semantic Playbook Lookup
Contextual **Accuracy**



Analysis Pipeline: Four-Phases Threat Intelligence Pipeline





PRIMARY UPLOAD

Ingest a file into Supabase

Pick a file, choose the source type, attach it to an org context, and watch the upload move into analysis.

Signed in as: villegasjoaco11@gmail.com

Database path: Supabase-backed analysis jobs

Log file

Choose File No file chosen

CSV, TXT, LOG, or JSON files up to your backend limit.

Source type

Firewall ▼

Stored in the backend as source_type and used for Supabase records.

Org context

1e62e598-f637-4b0a-854e-e67b12e22c88

Stored in the backend as org_id and used for Supabase records.

0%

Elapsed: 0s

ETA: estimating...

Ready to upload

Choose a log file to start.

0%

Start upload

Reset

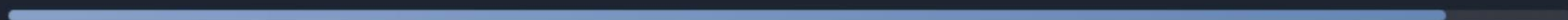
Recent Analysis

High

An incident involving a potential brute force attack was detected originating from the internal IP address 172.16.0.5 targeting multiple ports on the destination IP 192.168.50.1. The logs indicate a high frequency of connections within a short time frame, suggesting an attempt to compromise the system through repeated login attempts. Additionally, there were indications of anomalous traffic patterns, including unusually long flow durations and high packet counts, which deviate from normal operational behavior. This incident has been classified as high severity due to the nature of the brute force attempts and the potential risk to the targeted system's integrity and availability.

Threats found: 19 • MITRE mappings: 2

Prediction confidence: 92%



CONFIDENCE BREAKDOWN

LLM signal: 85%

MITRE retrieval: 97%

RF risk: 73%

Correlation evidence: 88%

Consistency bonus: 7%

Latest job status

Status: completed

Progress: 100%

Top Findings

Brute force attempts

Multiple connections from the same source IP to various destination ports within a short time frame, indicating potenti...

High

Suspicious IPs

Connections to external IPs that are not part of the internal network, indicating potential reconnaissance or data exfi...

Medium

Anomalous patterns

Unusual flow durations and packet counts that deviate from normal traffic patterns.

Medium

MITRE Techniques

Network Denial of Service

T1498 • impact

71%

Brute Force

T1110 • Unknown tactic

0%

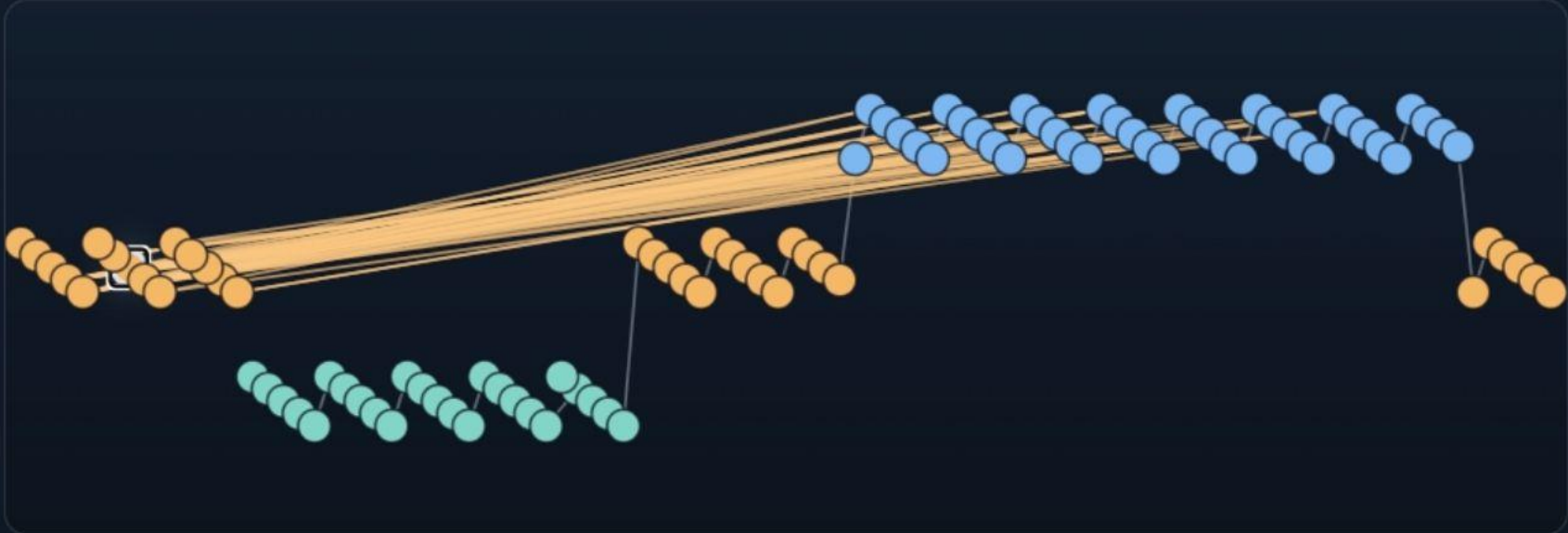
Relationship Graph

Events: 40

Detections: 35

AI Narrative: 25

Edges: 132



SELECTED NODE

Clear

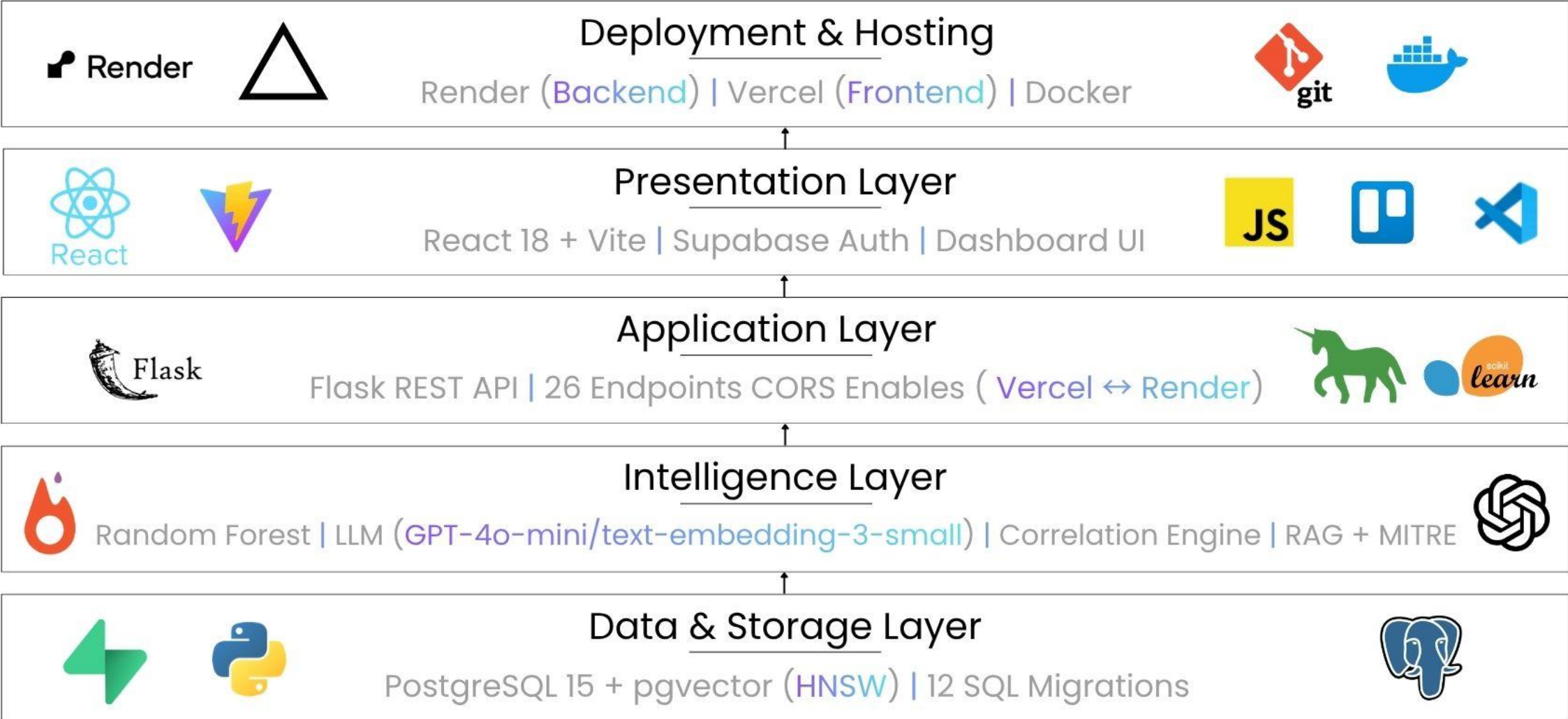
Detection

Existence rule triggered: 195 matching events found | Severity: high

NODE ID detection::be603efe-4994-4ef5-a8ac-e39627a7c367	TYPE Detection
SEVERITY high	TIMESTAMP Unknown time
SOURCE FILE e3303541-7ded-487e-b892-722a823d1212	CONNECTED EDGES 0

● Events ● Detections ● AI Narrative

Architecture & Tech Stack:



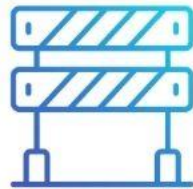
Challenges & Project Constraints:

Development Constraints

Solo Developer Project
Time-Bound (**1 Semester**)

Free-tiers

Security Audit Scope



Technical Scope Limits

Batch Processing Only
Cloud LLM **Dependency**

Rule-Based Correlation

Limited **Integrations**



Model Performance & Confidence Scoring

Training Foundation

CICIDS2017 & 2019
25+ Attack Classes
5-Fold Stratified CV



Confidence Tuning

Low (<0.5) → Downgrade
Medium ($0.5-0.8$) → Standard
High (>0.8) → Escalate



Key Metrics for Experts

Weighted Precision Target: ≥ 0.80 | Calibration: Sigmoid (Platt Scaling)

Detection Coverage: Intelligence at a Glance

AI & ML Power

150 Decision Trees (RF)
1,536 Dimension Vector Search
0.80 Precision Threshold

Threat Knowledge Base

25+ Attack Classes Mapped
8 DDoS Variants
2 Credential Attack Types
4 Web/DoS Exploit Patterns

System Performance

26 REST API Endpoints
110MB/2GB File Upload Capacity
~1 Sec Parse/Classification Latency

Security Posture

Row-Level Security (RLS)
Role-Based Access (RBAC)
Multi-Tenant Isolation

Future Enhancements:

Next

Real-Time
Streaming



WebSocket



Planned

Splunk
Integration



Bi-directional SIEM

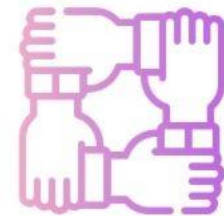


Vision

Enhanced
Correlation



Cross-File & Cross-
org pattern hunting



Conclusion:

Combining ML classification with RAG-based threat intelligence reduces time-to-detection and surfaces attack patterns that manual analysis misses.

Key Achievements

Semantic MITRE Mapping – Not just keyword matching

Works end-to-end – Not just a prototype

Reduced MTTD & MTTR

Q&A

THANK YOU!

References:

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WEF. (2025). *Global Cybersecurity Outlook 2025*.

NIST. (2024). *Incident Response Recommendations... CSF 2.0 Community Profile*.

Rahman, M.R., et al. (2021). *What are the attackers doing now?*

Sharafaldin, I., et al. (2018). *Toward Generating a New Intrusion Detection Dataset... (CICIDS2017)*.

MITRE ATT&CK® Framework. (2025). *Enterprise Matrix*.

NIST SP 800-61 Rev. 3 (Detection Process)